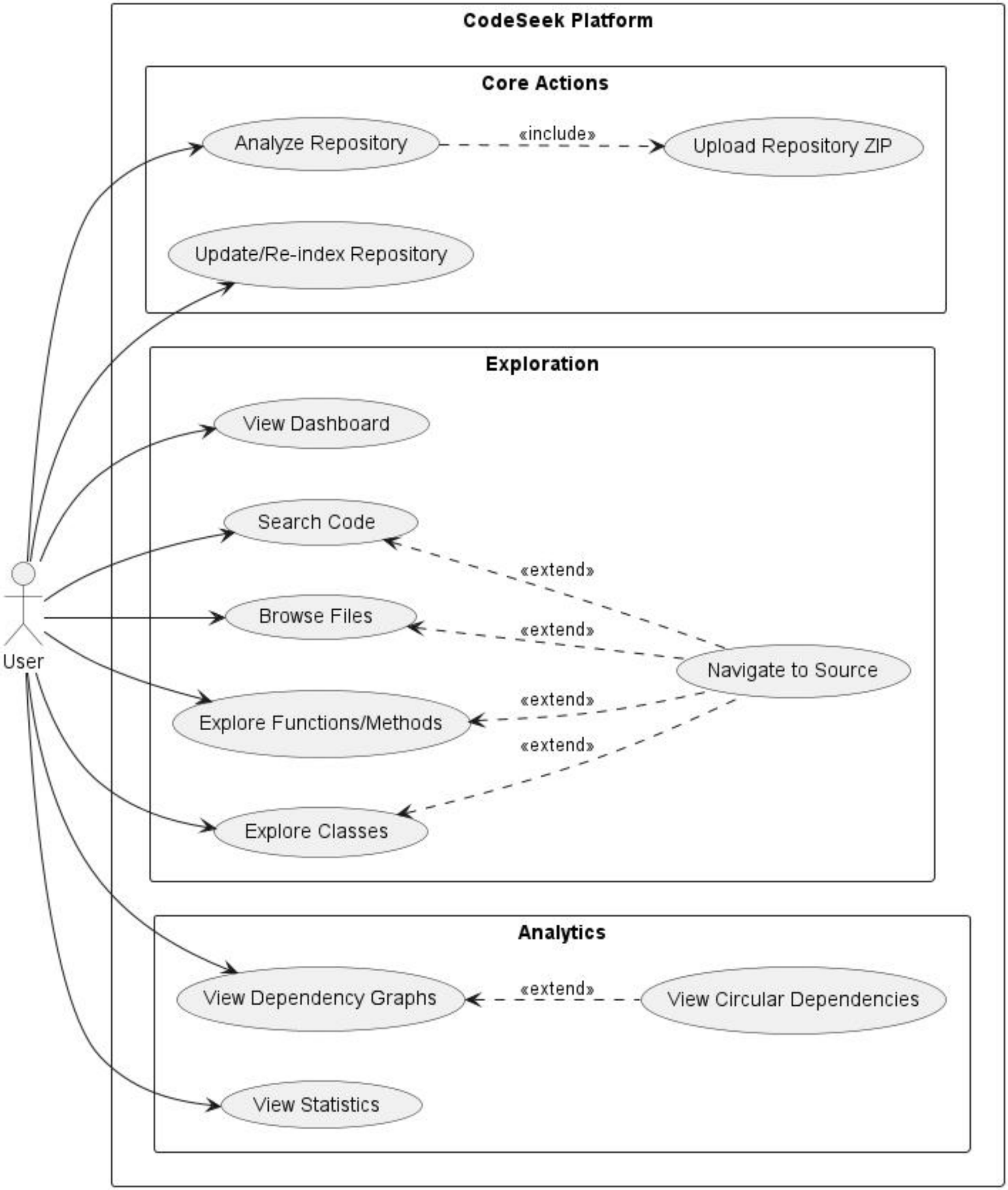
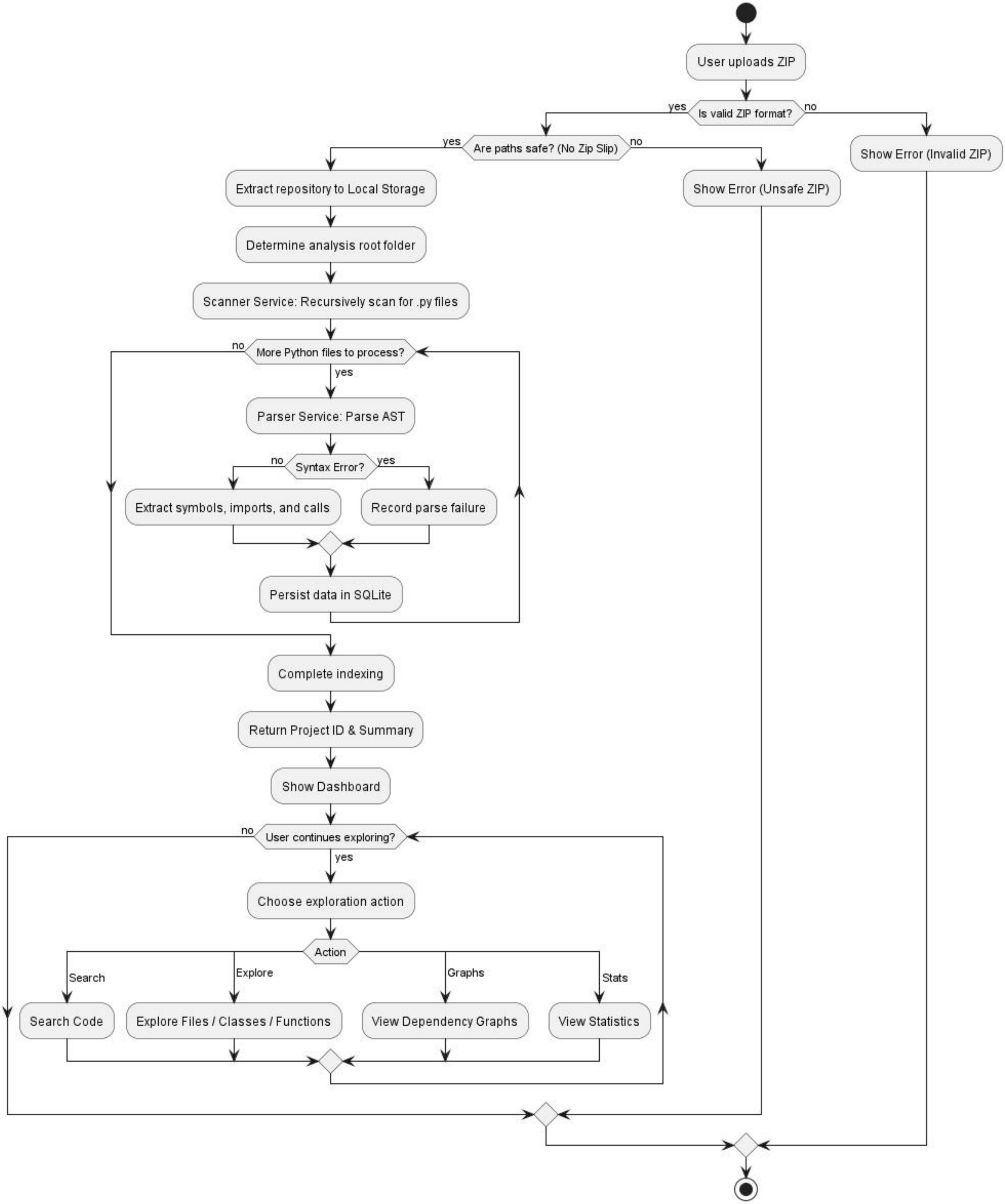
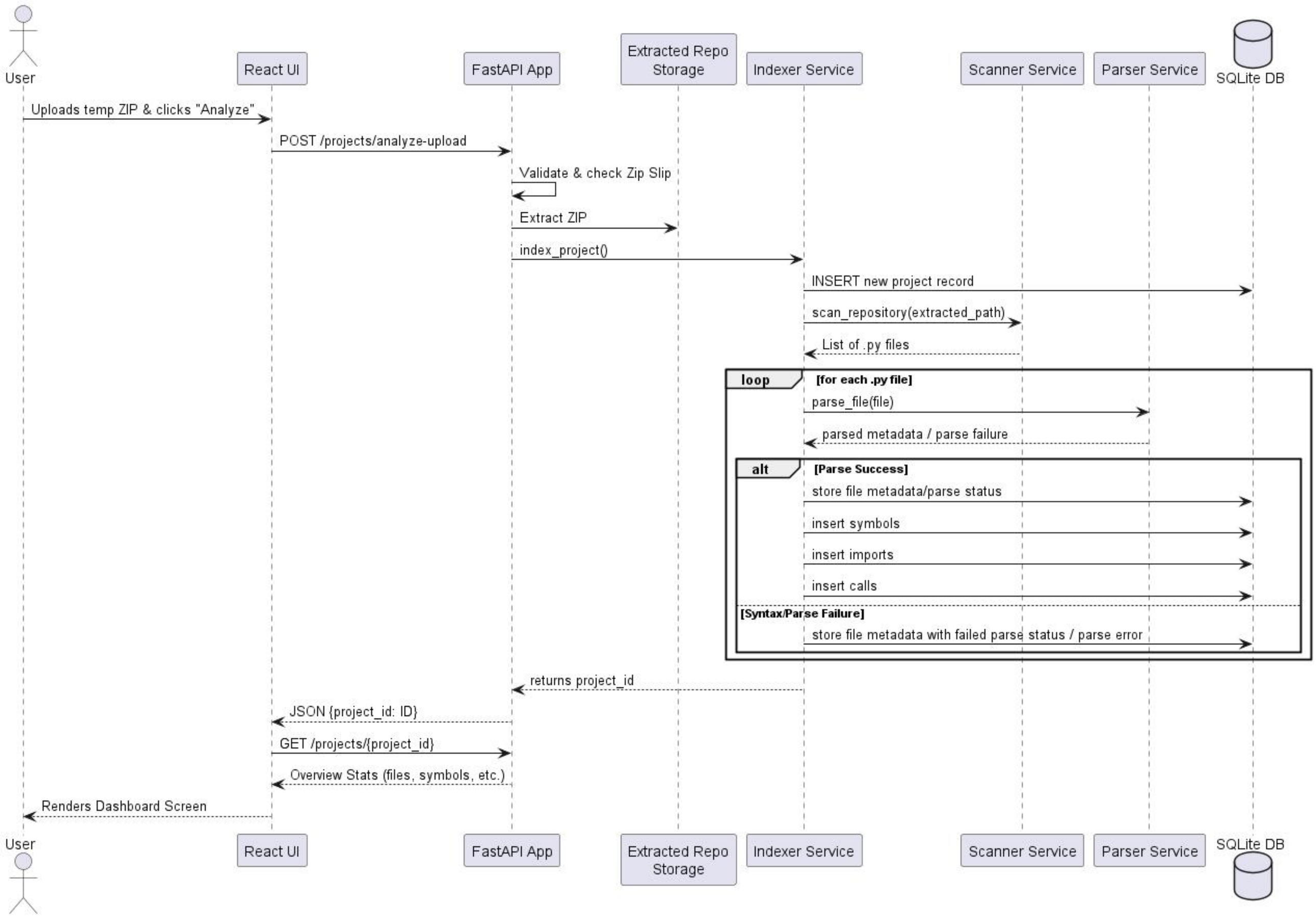
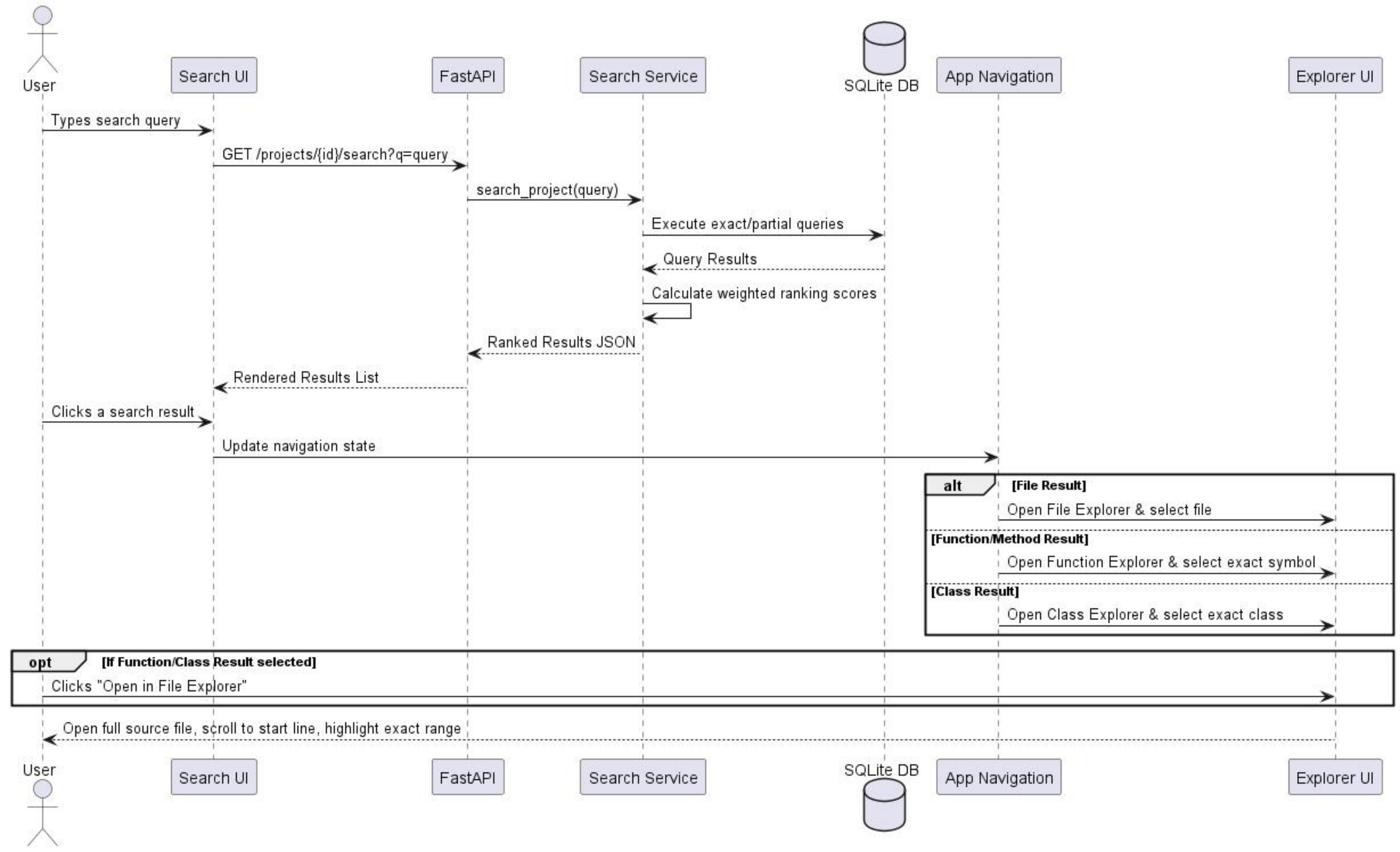
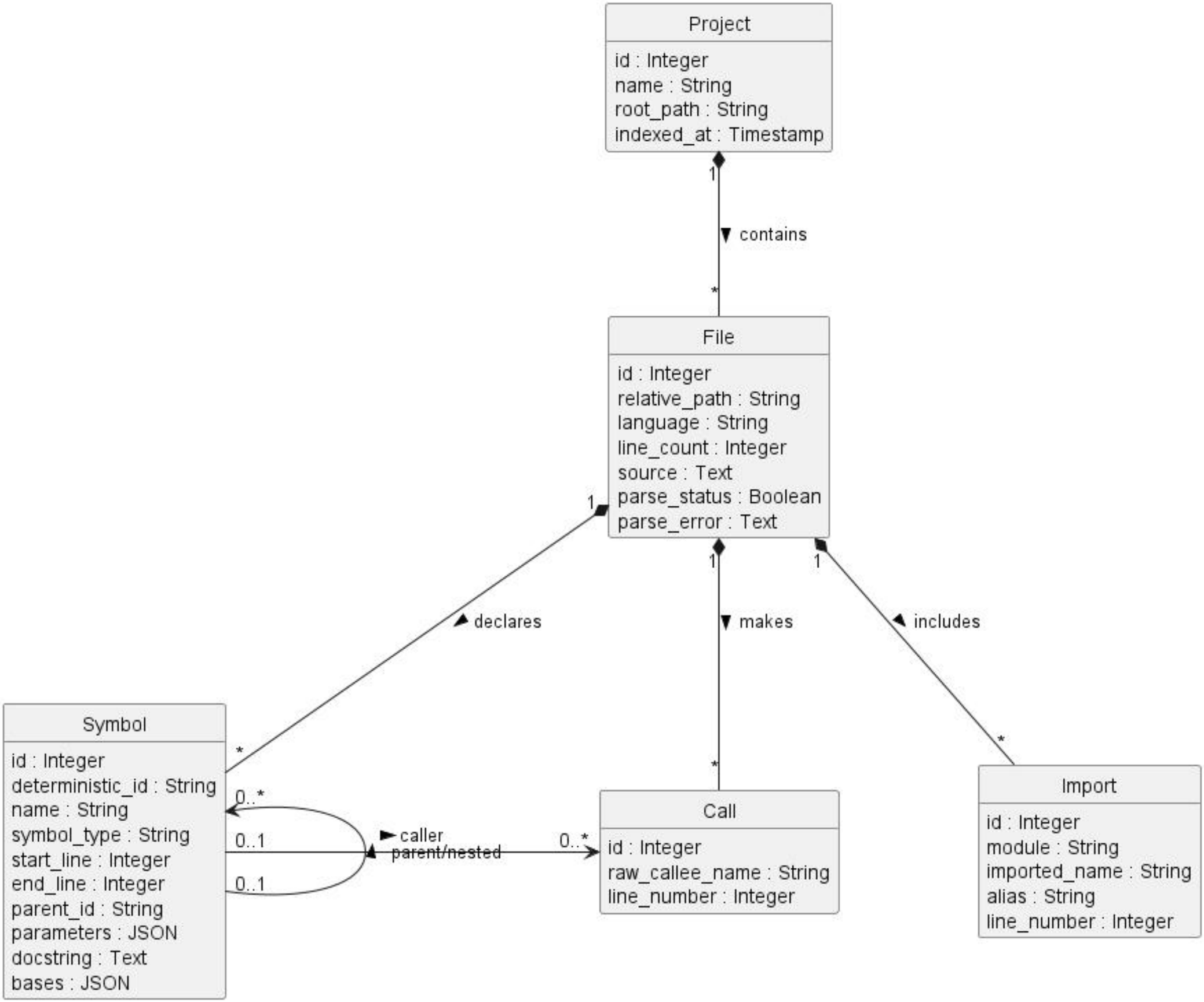
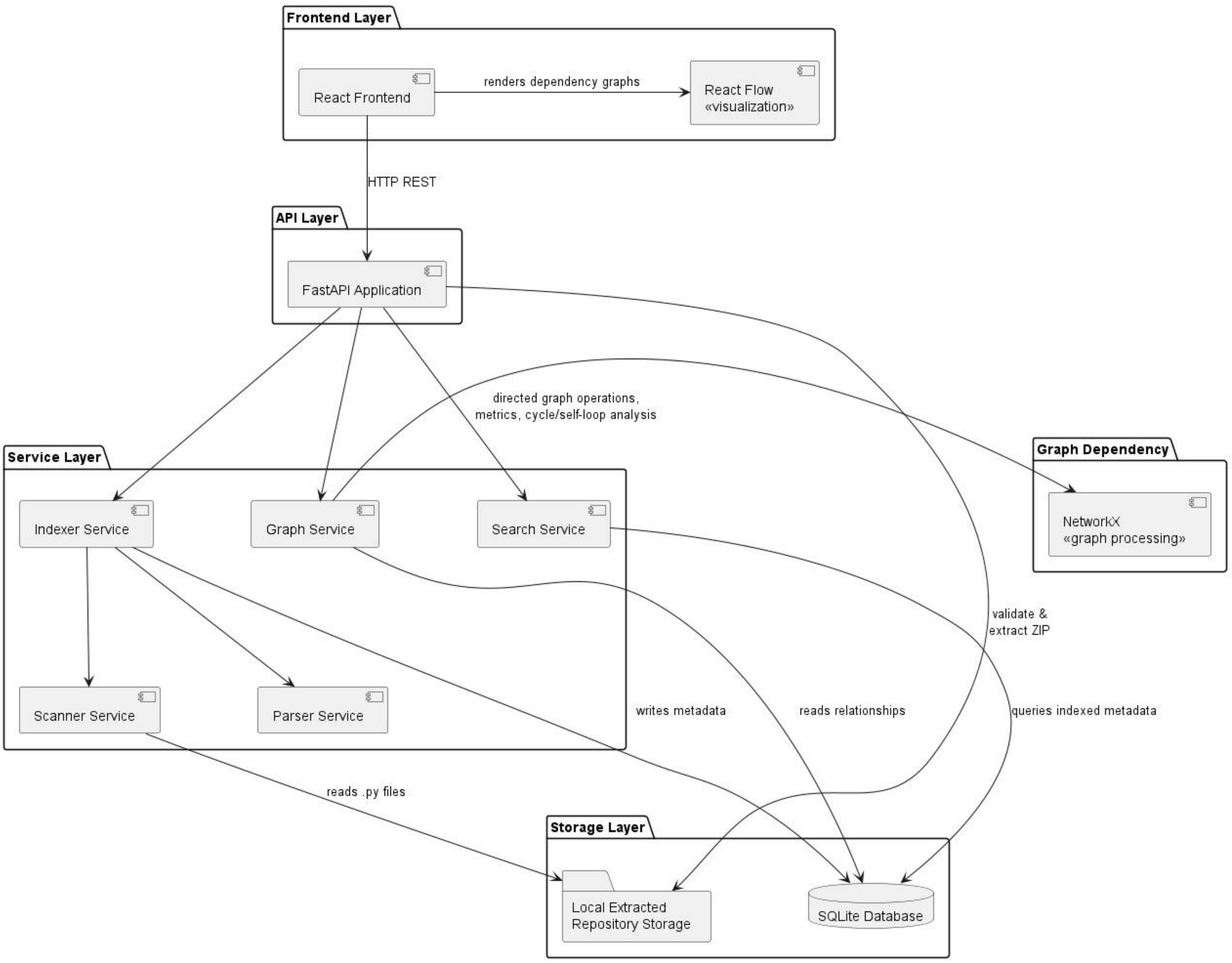

CodeSeek Platform
Core Actions
Analyze Repository
«include»
Upload Repository ZIP
Update/Re-index Repository
Exploration
View Dashboard
Search Code
«extend»
Browse Files
«extend»
Navigate to Source
Explore Functions/Methods
«extend»
«extend»
Explore Classes
User
Analytics
View Dependency Graphs
«extend»
View Circular Dependencies
View Statistics

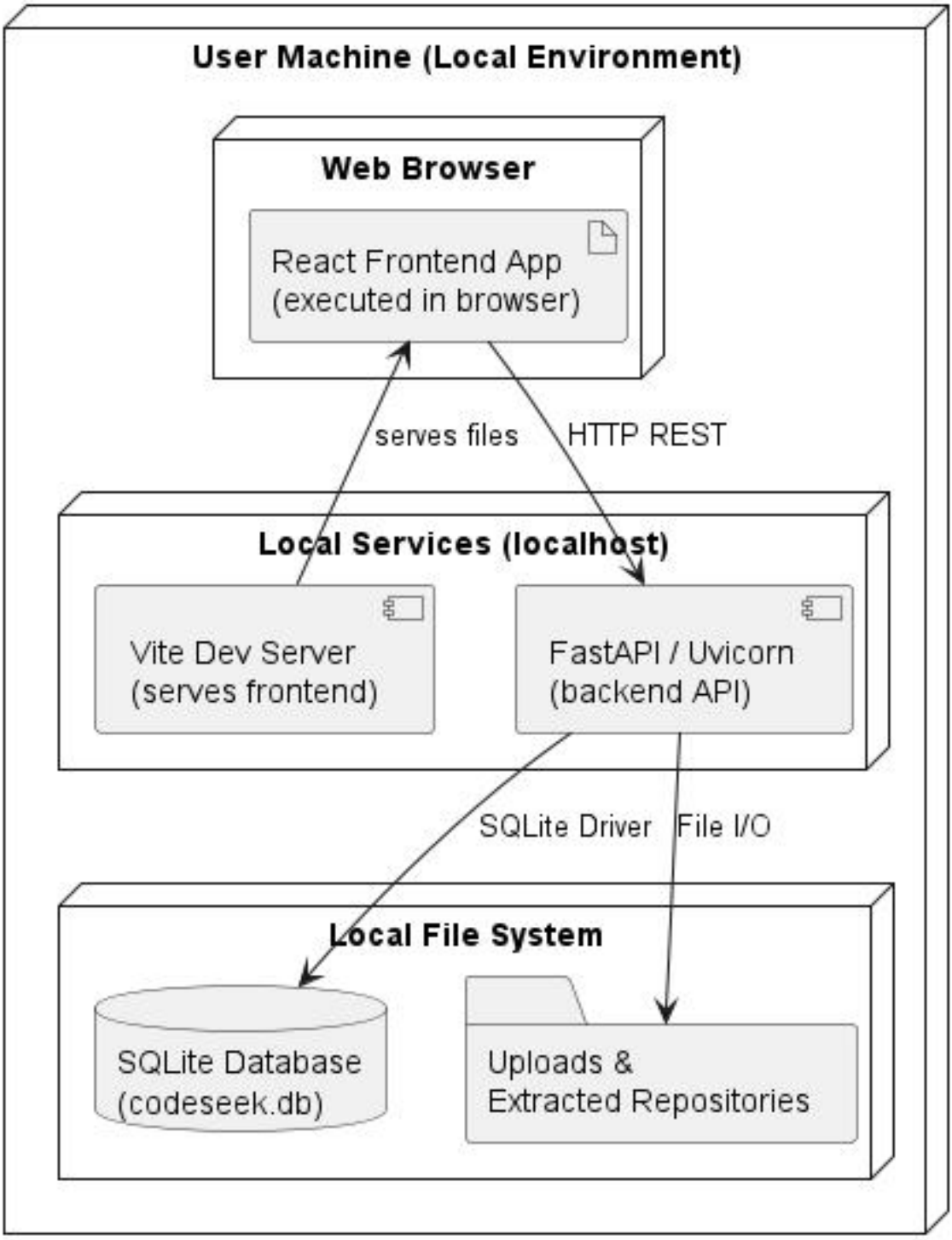

User uploads ZIP
Is valid ZIP format?
yes
no
Show Error (Invalid ZIP)
Are paths safe? (No Zip Slip)
yes
no
Show Error (Unsafe ZIP)
Extract repository to Local Storage
Determine analysis root folder
Scanner Service: Recursively scan for .py files
More Python files to process?
no
yes
Parser Service: Parse AST
Syntax Error?
no
yes
Extract symbols, imports, and calls
Record parse failure
Persist data in SQLite
Complete indexing
Return Project ID & Summary
Show Dashboard
User continues exploring?
no
yes
Choose exploration action
Action
Search
Explore
Graphs
Stats
Search Code
Explore Files / Classes / Functions
View Dependency Graphs
View Statistics


User
React UI
FastAPI App
Extracted Repo Storage
Indexer Service
Scanner Service
Parser Service
SQLite DB
Uploads temp ZIP & clicks "Analyze"
POST /projects/analyze-upload
Validate & check Zip Slip
Extract ZIP
index_project()
INSERT new project record
scan_repository(extracted_path)
List of .py files
loop
[for each .py file]
parse_file(file)
parsed metadata / parse failure
alt
[Parse Success]
store file metadata/parse status
insert symbols
insert imports
insert calls
[Syntax/Parse Failure]
store file metadata with failed parse status / parse error
returns project_id
JSON {project_id: ID}
GET /projects/{project_id}
Overview Stats (files, symbols, etc.)
Renders Dashboard Screen
User
React UI
FastAPI App
Extracted Repo Storage
Indexer Service
Scanner Service
Parser Service
SQLite DB


User
Search UI
FastAPI
Search Service
SQLite DB
App Navigation
Explorer UI
Types search query
GET /projects/{id}/search?q=query
search_project(query)
Execute exact/partial queries
Query Results
Calculate weighted ranking scores
Ranked Results JSON
Rendered Results List
Clicks a search result
Update navigation state
alt
[File Result]
Open File Explorer & select file
[Function/Method Result]
Open Function Explorer & select exact symbol
[Class Result]
Open Class Explorer & select exact class
opt
[If Function/Class Result selected]
Clicks "Open in File Explorer"
Open full source file, scroll to start line, highlight exact range
User
Search UI
FastAPI
Search Service
SQLite DB
App Navigation
Explorer UI


Project
id : Integer
name : String
root_path : String
indexed_at : Timestamp
1
contains
*
File
id : Integer
relative_path : String
language : String
line_count : Integer
source : Text
parse_status : Boolean
parse_error : Text
1
declares
*
1
makes
*
1
includes
*
Symbol
id : Integer
deterministic_id : String
name : String
symbol_type : String
start_line : Integer
end_line : Integer
parent_id : String
parameters : JSON
docstring : Text
bases : JSON
0..*
0..1
0..1
caller
parent/nested
0..*
Call
id : Integer
raw_callee_name : String
line_number : Integer
Import
id : Integer
module : String
imported_name : String
alias : String
line_number : Integer


Frontend Layer
React Frontend
renders dependency graphs
React Flow
«visualization»
HTTP REST
API Layer
FastAPI Application
directed graph operations,
metrics, cycle/self-loop analysis
Graph Dependency
NetworkX
«graph processing»
Service Layer
Indexer Service
Graph Service
Search Service
validate &
extract ZIP
Scanner Service
Parser Service
writes metadata
reads relationships
queries indexed metadata
reads .py files
Storage Layer
Local Extracted
Repository Storage
SQLite Database


User Machine (Local Environment)
Web Browser
React Frontend App
(executed in browser)
serves files
HTTP REST
Local Services (localhost)
Vite Dev Server
(serves frontend)
FastAPI / Uvicorn
(backend API)
SQLite Driver
File I/O
Local File System
SQLite Database
(codeseek.db)
Uploads &
Extracted Repositories